Assessment Task:

You are a data analyst at a garment manufacturing company. The company is interested in improving productivity on the factory floor. Your task is to provide data-driven insights to support their decision-making process.

You have a dataset on Moodle page named "garments_worker_productivity.csv". Please find below the variables' description:

- department : Associated department with the instance
- team_no : Associated team number with the instance
- no_of_workers : Number of workers in each team
- no_of_style_change : Number of changes in the style of a particular product
- targeted_productivity : Targeted productivity set by the Authority for each team for each day.
- smv : Standard Minute Value, it is the allocated time for a task
- wip : Work in progress. Includes the number of unfinished items for products
- over_time : Represents the amount of overtime by each team in minutes
- incentive : Represents the amount of financial incentive (in BDT) that enables or motivates a particular course of action.
- idle_time : The amount of time when the production was interrupted due to several reasons
- idle_men : The number of workers who were idle due to production interruption
- actual_productivity : The actual % of productivity that was delivered by the workers. It ranges from 0-1.

Assessment Details

<u>Inferential Statistics</u>

The company believes implemented new machines, and they believe this would reduce the average overtime (in minutes) that the workers will do, which will lead to a cost reduction in the end. They tried it in a group of 200 workers, which shows an average overtime of 6300 minutes. Can we say that the company is right? Formulate an appropriate hypothesis and justify your answer.

Formulate an appropriate analysis to determine the real average productivity of these workers each day of the week. Is there any difference between the days? Along with your results, provide a possible explanation on the outcome.

<u>Regression Analysis</u>

Use some variables of the dataset to perform regression analysis. You should identify the key variables that influence employee's productivity.

<u>Reporting</u>

Create a presentation summarizing your analysis above and findings. Explain limitations (if any) and suggest
further analysis to be performed. You are allowed to use descriptive statistics and visualisations to describe
the data. Your report should contain an introduction, body and conclusion. The length must be between
1000 and 1700 words.

1. Introduction

This report presents a comprehensive statistical analysis of garment factory worker productivity using the dataset *garments_worker_productivity.csv*. The company recently introduced new machinery with the expectation that it would reduce overtime and improve operational efficiency. As a data analyst, the objective is to evaluate this claim using inferential statistics, examine productivity patterns across days of the week, and identify key factors influencing actual productivity through regression modelling.

The analysis is structured into four main components:

1. **Hypothesis testing** to determine whether the new machines significantly reduced overtime.
2. **Productivity analysis across days of the week**, including estimation of real average productivity and statistical comparison.
3. **Regression analysis** to identify the variables that most strongly influence actual productivity.
4. **Interpretation, limitations, and recommendations** for further analysis.

The report uses descriptive statistics, visualisations, and inferential methods to provide data-driven insights that can support managerial decision-making in the garment manufacturing environment.

2. Dataset Overview and Preparation

The dataset contains **691 observations** and **13 variables**, including operational metrics such as overtime, work-in-progress (WIP), idle time, incentives, team size, and actual productivity. All variables are complete, with **no missing values**, which simplifies preprocessing and ensures that no imputation is required.

```
df=pd.read_csv('garments_worker_productivity.csv')
```

```
df.head()
```

	department	day	team	targeted_productivity	smv	wip	over_time	incentive	idle_time	idle_men	no_of_style_change	no_of_workers	actual_productivity
0	sweing	Thursday	8	0.8	26.16	1108	7080	98	0.0	0	0	59.0	0.940725
1	sweing	Thursday	11	0.8	11.41	968	3660	50	0.0	0	0	30.5	0.800570
2	sweing	Thursday	12	0.8	11.41	968	3660	50	0.0	0	0	30.5	0.800570
3	sweing	Thursday	6	0.8	25.90	1170	1920	50	0.0	0	0	56.0	0.800382
4	sweing	Thursday	7	0.8	25.90	984	6720	38	0.0	0	0	56.0	0.800125

```
df.shape
```

```
(691, 13)
```

2.1 Data Types and Structure

- **Numerical variables (11):** smv, wip, over_time, idle_time, incentive, actual_productivity, etc.
- **Categorical variables (2):** department, day

The dataset reflects real operational variability, including high overtime values, fluctuating WIP, and occasional production interruptions

```
df.dtypes.value_counts()
```

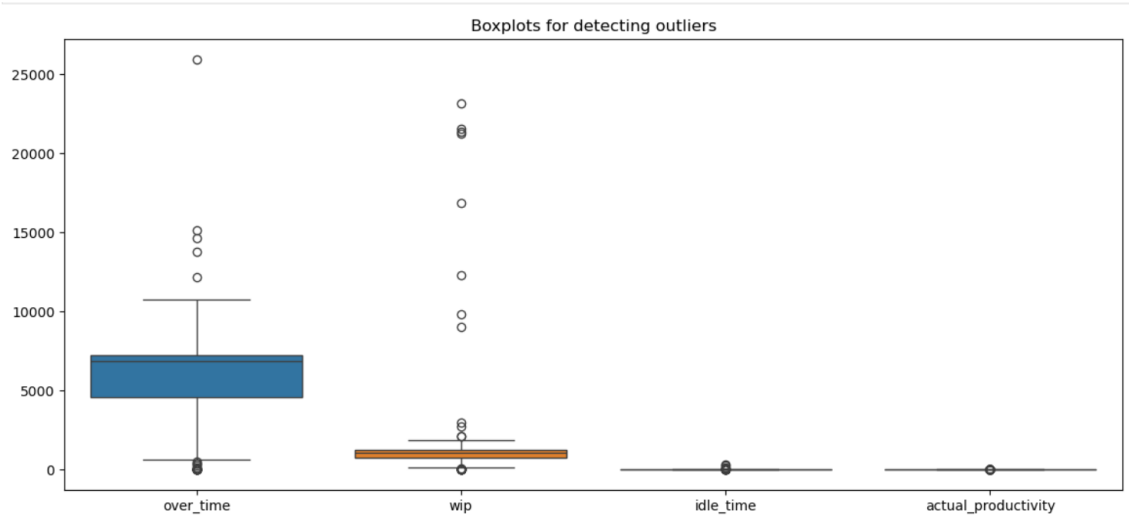
```
int64      6
float64    5
object     2
Name: count, dtype: int64
```

2.2 Outlier Detection

Outliers were assessed using boxplots and Z-score analysis. A total of **97 extreme values** were identified, particularly in:

- **over_time**
- **wip**
- **idle_time**

These outliers appear to represent genuine operational peaks rather than data errors. Removing them would distort the true production environment, so they were retained. Their presence is acknowledged as a factor that may increase variance in statistical tests.




```
numeric_cols = df.select_dtypes(include=['float64', 'int64']).columns
z_scores = np.abs(zscore(df[numeric_cols]))

outliers_count = (z_scores > 3).sum()
outliers_count
```

```
np.int64(97)
```

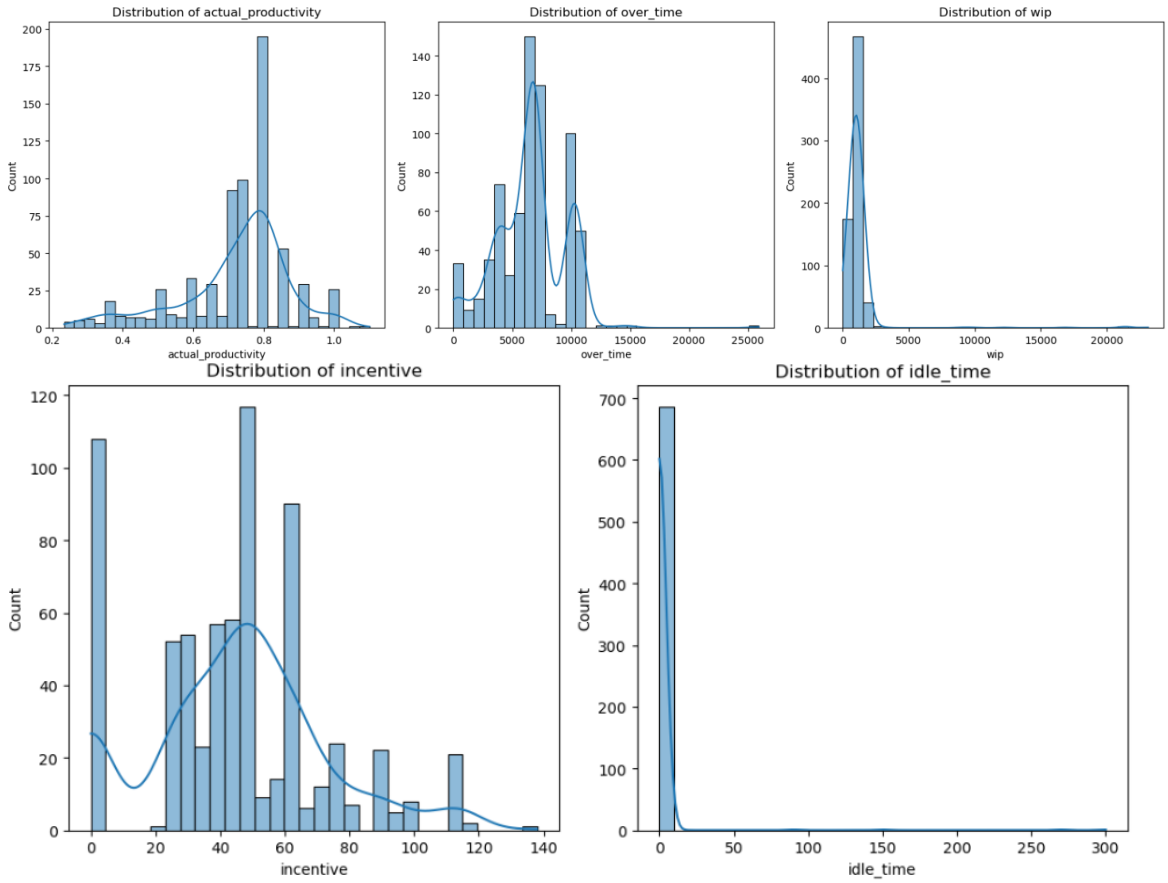
3. Exploratory Data Analysis (EDA)

3.1 Distribution of Key Variables

Visual inspection of histograms reveals:

- **Actual productivity** is concentrated between **0.7 and 0.9**, indicating generally high performance.
- **Overtime** is heavily right-skewed, with a few extreme days exceeding 20,000 minutes.
- **WIP** shows a long right tail, suggesting occasional bottlenecks.
- **Incentives** are often zero, indicating that financial rewards are not consistently applied.
- **Idle time** has many zero values but some large spikes, consistent with rare production interruptions.

These patterns highlight the operational complexity of garment manufacturing and the need for robust statistical methods.

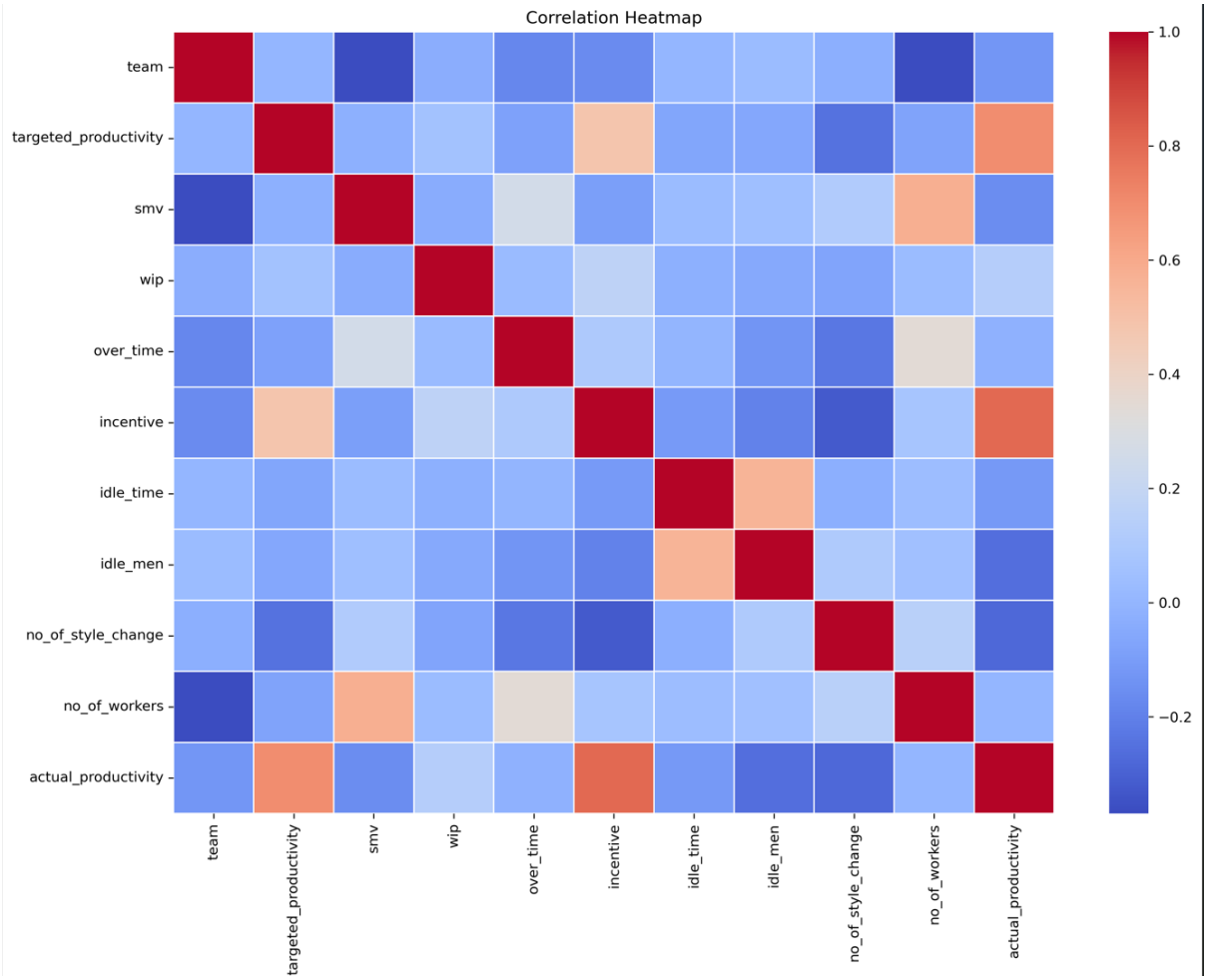


3.2 Correlation Analysis

A correlation heatmap shows several important relationships:

- targeted_productivity has a strong positive correlation with actual_productivity, suggesting that higher planned targets are associated with higher delivered productivity.
- over_time has a weak negative correlation with productivity, indicating diminishing returns when overtime becomes excessive.
- idle_time and idle_men are strongly correlated, as expected.
- wip correlates moderately with over_time, implying that high WIP may lead to longer working hours.

These insights guide variable selection for regression modelling.

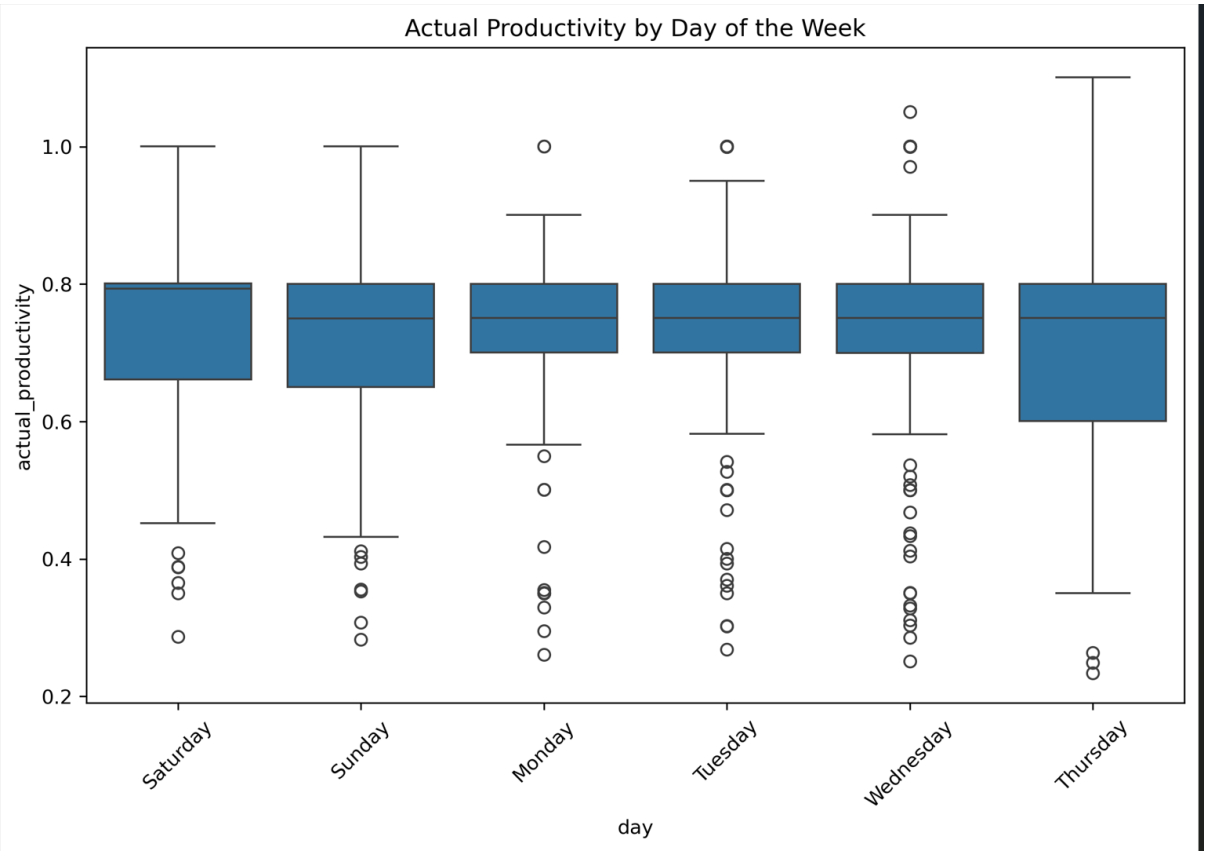


3.3 Productivity by Day of the Week

A boxplot of actual productivity by day shows:

- Slight variations in median productivity across days.
- Saturday appears to have slightly higher productivity.
- Thursday appears to have slightly lower productivity.

However, visual differences alone do not confirm statistical significance, motivating the use of ANOVA.



5. Inferential Statistics

4.1 Hypothesis Test: Did the New Machines Reduce Overtime?

The company tested new machines on 200 workers, observing an average overtime of 6,300 minutes. To evaluate whether this represents a significant reduction, we compare it to the historical mean overtime from the dataset.

- Historical mean overtime: 6,508 minutes
- Standard deviation: 2,864 minutes
- New sample mean: 6,300 minutes
- Sample size: 200 workers

Hypotheses

- H_0: μ_new = μ_old (no reduction in overtime)
- H_1: μ_new < μ_old (overtime decreased)

A one-sample left-tailed t-test was performed.

Results

- t-statistic: –1.03
- p-value: 0.152

Interpretation

Since $p > 0.05$, we fail to reject the null hypothesis.

There is not enough statistical evidence to conclude that the new machines significantly reduced overtime. Although the new mean is lower, the difference is not statistically meaningful given the variability in the data.

This suggests that:

- The machines may not yet be producing measurable efficiency gains.
- The sample of 200 workers may not fully represent the broader workforce.
- Operational factors beyond machinery may influence overtime

4.2 Productivity Differences Across Days of the Week

To determine whether productivity varies by day, we calculated descriptive statistics and performed a one-way ANOVA.
Mean Productivity by Day

	day	mean	std	count
0	Saturday	0.734080	0.144047	104
1	Sunday	0.713549	0.149421	116
2	Monday	0.724169	0.147301	116
3	Tuesday	0.728115	0.151940	118
4	Wednesday	0.722712	0.165911	119
5	Thursday	0.710772	0.168925	118

The differences are small, with Saturday slightly higher and Thursday slightly lower.
ANOVA Results
• F-statistic: 0.360
• p-value: 0.876
Interpretation
Since $p > 0.05$, we fail to reject the null hypothesis.
There is no statistically significant difference in average productivity across days of the week.
Possible explanations:
• Workload and staffing are consistent across days.
• Production processes are standardised.
• Incentives and targets do not vary significantly by day.

5. Regression Analysis
The goal of the regression model is to identify which variables most strongly influence actual productivity.

5.1 Model Setup
• Dependent variable: actual_productivity
• Independent variables: targeted_productivity, smv, wip, over_time, incentive, idle_time, idle_men, no_of_style_change, no_of_workers, and day dummies.
A multiple linear regression model was fitted using an 80/20 train-test split.

5.2 Model Performance

• R-squared: 0.802
• RMSE: 0.065
This indicates that the model explains approximately 80% of the variance in productivity, which is strong for operational data.

5.3 Key Predictors

The most influential variables (based on coefficients and statistical significance) are:
Strong Positive Predictors

• targeted_productivity (largest positive effect)
• incentive
• no_of_workers
• idle_time (small positive effect)
Negative Predictors
• over_time
• idle_men
• smv
Non-significant Predictors
• wip
• no_of_style_change
• day variables
Interpretation
• Targeted productivity is the strongest driver of actual productivity. Higher expectations appear to motivate higher output.
• Incentives significantly increase productivity, confirming the effectiveness of financial motivation.
• Overtime reduces productivity, likely due to fatigue.
• Idle men strongly reduces productivity, reflecting operational inefficiencies.
• Day of the week has no meaningful effect, consistent with ANOVA results.

6. Limitations

While the analysis provides valuable insights, several limitations should be acknowledged:

6.1 Outliers

Extreme values in overtime and WIP may influence statistical tests and regression coefficients. Although they appear genuine, they increase variance and reduce statistical power.
6.2 Missing Contextual Variables
The dataset lacks information on:
• Worker experience
• Machine type and age
• Shift duration
• Product complexity
These factors may influence productivity but are not captured.

6.3 Causality Cannot Be Proven

Regression identifies associations, not causal relationships. For example, higher targeted productivity may correlate with better teams rather than cause higher productivity.

6.4 Limited Experimental Control

The new machine test was not a controlled experiment. Without random assignment or pre-post data, conclusions about machine impact remain tentative.

7. Recommendations for Further Analysis

To strengthen insights and support more robust decision-making, the following analyses are recommended:

7.1 Pre-Post Intervention Study

Collect overtime data before and after machine installation for the same teams. This would allow:
• Paired t-tests

• Difference-in-differences analysis
• Time-series modelling

7.2 Productivity Forecasting

Use machine learning models (e.g., Random Forest, Gradient Boosting) to predict productivity and identify nonlinear relationships.
7.3 Operational Bottleneck Analysis
Investigate:
• WIP spikes
• Idle time events
• Team-level performance differences

7.4 Worker-Level Data Collection

Adding demographic and skill-level data would enable more granular modelling.

7.5 Cost-Benefit Analysis

Quantify:
• Cost of new machines
• Savings from reduced overtime
• Productivity gains
This would support strategic investment decisions.

8. Conclusion

This analysis provides a detailed examination of garment worker productivity using descriptive statistics, inferential methods, and regression modelling. The key findings are:
1. The new machines did not significantly reduce overtime, based on the available sample.
2. Productivity does not differ significantly across days of the week, indicating stable operational performance.
3. Targeted productivity, incentives, and team size are the strongest positive predictors of actual productivity.
4. Overtime, idle men, and SMV negatively affect productivity, highlighting areas for operational improvement.
While the dataset offers valuable insights, further data collection and more controlled experimental designs would enhance the reliability of conclusions. Nevertheless, the results provide a strong foundation for managerial decisions aimed at improving efficiency, reducing overtime, and optimising workforce performance.

CCT College Dublin (2025) *garments_worker_productivity.csv* [Dataset]. Provided via Moodle for CA2 assessment.